
Where does the VAR prior fail?

Block-adaptive Bayesian local projections
under sparse dynamic misspecification

MSc thesis project Stefano Graziosi 12 September 2026 Status: methodological
prototype, $R = 500$ Monte Carlo evidence, and a first pass at the euro-area application.

The research question

Local projections

Estimate each horizon directly. They tolerate richer dynamics, but impulse responses can be imprecise in modest samples.

VARs

Impose one dynamic system across horizons. They gain precision when that system is adequate, but transmit misspecification everywhere.

A global Bayesian LP compromises by shrinking every coefficient block toward a VAR-implied centre with the same horizon-specific tightness λ_h .

When the VAR is wrong in one part of the system, can only that part of the LP escape the prior, and does it help?

Block-adaptive VAR-centred shrinkage

$$y_{i,t+h} = z_t' \beta_{i,h} + u_{i,t+h},$$
$$\beta_{i,g,h} - \mu_{i,g,h}^{\text{BVAR}} \sim \mathcal{N}(0, \sigma_{i,h}^2 \lambda_h^2 \tau_{i,g,h}^2 D_{g,h}), \quad \tau_{i,g,h} \sim \mathcal{C}^+(0, 1).$$

In each response equation and horizon, block g collects the p coefficients on the lags of predictor g . Three estimators differ only in how τ is treated:

BLP-FMAR $\tau \equiv 1$: the published global baseline

BLP-block $\tau_{i,g,h}$ independent at every horizon

BLP-pooled random walk on $\log \tau_{i,g,h}$ across h

$\tau_{i,g,h} < 1$ tighter than the global prior

$\tau_{i,g,h} = 1$ the global FMAR baseline

$\tau_{i,g,h} > 1$ local escape from the VAR

Shrinkage acts on deviations from the BVAR centre, not toward zero.

With every $\tau = 1$ both adaptive estimators reproduce the FMAR closed form to 3×10^{-16} (912 cells). Baseline: Ferreira, Miranda-Agrippino, Ricco (2025); sampler: Makalic and Schmidt (2016).

A deliberately diagnostic Monte Carlo design

Regime	True dynamics	What the design tests
Correct	Stable VAR(2)	Does unnecessary local flexibility cost precision?
Sparse	VAR(3), only $A_3(3, 1) = -0.30$ missing from the fitted VAR(2)	Does block 1 escape in the y_3 equation, where the transmission prior is wrong?
Intermediate Dense	VAR(3), nonzero first <i>column</i> of A_3 VARMA(2,1), dense MA matrix	The same block wrong in every equation Does the advantage vanish when no block is responsible?

Headline experiment. $K = 3$, $T = 200$, fitted $p = 2$, $H = 20$; $R = 500$ (correct, sparse), $R = 250$

(intermediate, dense); a recursive unit shock to y_1 . **Comparison.** LP, BVAR, BLP-FMAR, BLP-block, BLP-pooled on the same samples; paired Monte Carlo t statistics; integrated RMSE over $h \geq 2$; τ localisation against a measured null.

What is implemented now

Method

FMAR BVAR and global BLP; block-adaptive Gibbs sampler; horizon-pooled sampler (checkerboard Metropolis on $\log \tau$); HAC bands; Rao–Blackwellised point estimates.

Fair benchmark

$h = 1$ estimated as an LP for every estimator; integrated RMSE over $h = 2..H$; common λ_h ; paired comparisons on identical samples.

Evidence

Five-estimator headline runs for all four DGPs, a 13-cell exploratory grid, a sensitivity analysis of the remaining approximations; every table regenerated from stored files.

Verification

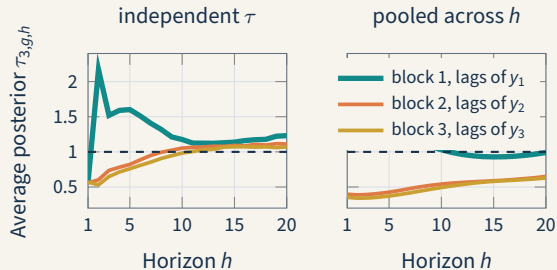
17 assertion tests pass in MATLAB R2025b (12 minutes): exact nesting, chunk-exact parallel runs, the prior-scale floor, the held-block option, the empirical pipeline.

Empirical

Euro-area data assembled and validated; the full pipeline has run (five estimators, dose–response in p , JK split, coherence, null at $R = 500$); three corrections were needed on the real data.

Caveats. The fidelity test against the original FMAR code still skips (package not installed); the raw outcome series are third-party data, not in the repository.

The scales locate the failure; pooling makes them sharp



Sparse DGP, y_3 equation, 500 replications.

Localisation, early horizons. The true (equation, block) cell ranks first in **97%** of replications; **37%** on the correct DGP.

Detection. Some cell has $P(\tau > 1) > 0.5$ in **74%** of sparse replications, **3%** of correct ones.

		early	all $h \geq 2$	late
sparse	block	.968	1.012	1.040
	pooled	.973	1.003	1.021
correct	block	.965	1.011	1.031
	pooled	.949	.988	1.005

Paired RMSE ratio vs BLP-FMAR, < 1 = adaptation helps; $|t| > 3$ except the bold entries.

What the project can claim today

Established

A calibrated, localised diagnostic of prior-data conflict: the true cell is found 97% of the time against a 37% measured null; an honest *no* on the dense DGP; monotone in conflict strength; exactly nested in FMAR.

Established

Adaptation lowers RMSE early and raises it late on every DGP but one; pooling keeps the early gain and halves the late penalty, so the adaptive layer is RMSE-neutral.

Finding

Knowing *where* a VAR prior fails does not imply that escaping there helps: on the intermediate DGP the map localises perfectly and adaptation is worse at every horizon.

Not established

Integrated-RMSE superiority; a τ threshold that transfers across specifications (the null moved from 0.42 to 0.68 with the fitted lag order); any statement about euro-area transmission.

Framing the evidence supports: a specification diagnostic for VAR-centred local projections, with horizon pooling as the refinement that makes it sharp and RMSE-neutral.

Euro area: three things the real data forced

Monthly 2000–2019, $K = 5$: EA-EMPD 1-year OIS surprise (internal instrument, ordered first), 12-month Euribor, industrial production, HICP, EURO STOXX 50; $p = 12$, $H = 48$.

- Weak instrument** The surprise innovation moves the rate innovation with $t = 0.77$ ($F = 0.6$), not the $F \approx 4$ budgeted: a monthly-average indicator smears events over two months, and the largest surprises sit in late 2008 against 90 bp rate falls. The τ diagnostic does not depend on this; the IRFs and the 25 bp scale do.
- Prior scale** FMAR's Newey–West scale falls to a fifth of the variance for the near-white-noise surprise; the marginal likelihood answers by driving the *global* λ_h to 0.01, bimodal at 28 of 48 horizons. A floor at the residual variance (inert on the simulations) restores a smooth path.
- Instrument block** One large contemporaneous coefficient and eleven near-zero lags violate the block prior's common scale; freed, the block collapsed and pulled the identifying coefficient to the VAR centre. It is held at $\tau = 1$; the diagnostic is read on the four level blocks.

Each choice defaults to the published behaviour, is pinned by a test and documented in `empirical/README_EMPIRICAL.md`.

Euro area: the τ map and its null

Early-horizon mean scale $\bar{\tau}_{i,g}$, $h = 2..12$, pooled (independent in brackets); rows = equations, columns = level blocks.

	i1y	ip	hicp	stoxx
mps	0.07 (0.5)	0.10 (0.6)	0.05 (0.6)	0.09 (0.7)
i1y	4.6 (2.0)	2.4 (1.3)	0.9 (0.9)	2.7 (1.7)
ip	7.6 (5.3)	4.4 (3.2)	2.1 (1.9)	1.8 (1.3)
hicp	3.3 (2.4)	2.3 (1.8)	10.0 (9.4)	3.0 (2.3)
stoxx	4.4 (2.6)	3.3 (1.9)	0.4 (0.6)	0.3 (0.7)

Negative control passes (surprise equation quiet at every p); **coherence holds** (blocks with $\tau > 2$ closer to their unshrunk value than under FMAR in 100% of cells).

Sources: results/tau_protocol_p12.csv and empirical/docs/CH7_RESULTS.md.

Against the simulated null ($R = 500$ datasets from the fitted BVAR through the whole pipeline):

Equation-wide q95 of the largest $\bar{\tau}$ is **5.5–8.5**: the null is wide. Two cells survive: the **HICP own block** (9.4 vs 7.0; pooled 10.0 vs 8.3) and the **rate block in the IP equation** (pooled 7.6 vs 7.4). All rate-block cells in the i1y and stoxx equations are inside their null.

Through the protocol the dose–response in p is flat (1 / 1 / 2 escapes at $p = 2/6/12$), and in the level system without the surprise only the HICP block survives its own null.

Highest-return next steps

- 1 Treat the instrument the AGKL way.** Zero restrictions on the surprise block remove the drag on λ_h at the source instead of rescaling it; joint selection of λ_h and the block scales is the general version.
- 2 A stronger policy indicator.** An end-of-month 1-year OIS level instead of the monthly-average Euribor removes the two-month smearing and the 2008–2012 credit premium.
- 3 Robustness the design lists.** The 3-month surprise, the speeches-augmented series (all months covered), the $K = 7$ system with loans and spreads.
- 4 Method.** A non-centred parameterisation for the pooled $\log \tau$ chain (ESS 10–70 of 700); a factorial grid to see interactions between sparsity and sample size.

Best use of supervision: agree the diagnostic framing and the instrument treatment before writing the empirical chapter.

Restart map

Reorient

docs/PROJECT_EXPLAINED.md (the procedure and every file in plain language), then docs/RESULTS.md and empirical/docs/CH7_RESULTS.md.

Test

cd tests; run_all_tests (12 minutes in MATLAB). Set FMAR_PATH for the fidelity test against the original package.

Simulation evidence

report_montecarlo('results/mc_headline_sparse.mat');
results/REPORT.txt has every table.

Empirical chapter

SMOKE_TEST_EMPIRICAL, then RUN_EMPIRICAL (4 minutes) and run_null_calibration (85 minutes at $R = 500$); outputs carry the tags p12, p12_nofloor, p12_allblocks, _levels.

Method work

estimate_blp_blockpooled.m, gibbs_block_pooled_horizons.m, select_lambda_fmar.m, fmar_prior_scale.m.

First question on return: with the instrument block restricted the AGKL way, does the euro-area map keep the HICP and interest-rate blocks as its two features?